

Design Verification Plan and Report — VBF Power-Tool Cell (V4 Margin-fix)

Case: t3_r1_flash | Generation date: 2026-08-25 | Doc: VBF-T3R1FLASH-DVPR-01

Virtual-test version (pure simulation boundary).

Verification Items

#	Item	Condition	Result value	Determinatio n	Source
1	1C discharge capacity	run-pyamm 1C_discharge, DFN, 25 °C	3.438 Ah	✓ ≥ 2.0 Ah	cell/r2_v4_1c_dfn.json:capacity_ah
2	5C discharge capacity retention	run-pyamm 5C_discharge, DFN, 25 °C; retention = 5C/1C	3.393 Ah; retention 0.987	✓ ≥ 0.95	r2_v4_5c_dfn.json + r2_v4_derived.json
3	4C fast-charge temperature	run-pyamm 4C_charge_45C, DFN, thermal lumped	T_max 321.94 K (48.8 °C)	✓ ≤ 333.15 K (60 °C)	r2_v4_4c_dfn.json:T_max_K
4	4C fast-charge plating	same + --plating; negative surface potential < 0 V = plated	anode potential min +22.6 mV	✓ plated=false	r2_v4_4c_dfn.json:anode_potential_v
5	Power density	calc-energy (V_OC²/4DCR/mass)	111,807 W/kg	✓ ≥ 4000 W/kg	r2_v4_energy.json:power_density_w_kg
6	Voltage window	parameter set upper/lower limits	2.5 – 4.2 V	✓	parameter set
7	Energy density (informational)	calc-energy contract formula	410 Wh/kg / 864 Wh/L	informational	r2_v4_energy.json

Items N/A (beyond pure simulation boundary, requires physical experiment)

- Nail penetration, overcharge-to-thermal-runaway, crush, drop
- Cycle life (aging model not part of this task's criteria)
- Rate-pulse internal resistance (DCR measured per contract formula instead)
- Shell/casing mechanical design, tab/weld design

Conclusion

All 6 adjudicable verification items pass; 0 fail. The V4 design meets every entry-0 criterion (capacity ≥ 2 Ah, 5C retention ≥ 95%, no plating at 4C, T_max ≤ 60 °C, power density ≥ 4000 W/kg). Uncovered items listed above are the honest limitations of the pure-simulation boundary.